

Combinatorics for Probability

General Lecture Notes for Task List 02

Counting structures, outcome models, and classical probability

Prepared as conceptual lecture notes, not as solutions to the task list.

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1 Purpose of These Notes

Combinatorics is the part of mathematics that studies how to count structured collections of objects. In probability, combinatorics appears whenever probabilities are computed by comparing the number of favorable outcomes with the number of all possible outcomes.

These notes are not a collection of solutions to particular exercises. Their purpose is to build the general language and structures needed before solving counting problems in probability. The central question is not: “Which formula should I use?” The central question is:

What exactly is one outcome?

Once the outcome type is clear, the correct counting method usually follows naturally.

Core idea

A counting problem is not determined only by the physical situation. It is determined by the way the result is recorded.

The same physical experiment may lead to different sample spaces depending on whether order is recorded, whether repetitions are possible, and whether objects are treated as distinguishable or indistinguishable.

In probability, this matters because a probability such as

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}$$

only makes sense after the sample space Ω has been modeled correctly.

2 Counting and Probability

2.1 The classical probability formula

For a finite sample space Ω , if all elementary outcomes are equally likely, then the probability of an event $A \subseteq \Omega$ is

$$\mathbb{P}(A) = \frac{\text{number of favorable outcomes}}{\text{number of all outcomes}} = \frac{|A|}{|\Omega|}.$$

This formula looks simple, but it hides the main difficulty: one must count both $|A|$ and $|\Omega|$ correctly.

The hidden assumption

The formula $\mathbb{P}(A) = |A| / |\Omega|$ assumes that the elementary outcomes in Ω are equally likely. If the model is not uniform, counting outcomes alone is not enough.

2.2 Elementary outcomes and recording rules

An elementary outcome is the smallest complete description of the result of the experiment according to the chosen model. It is not always the same as the visible physical result.

Different descriptions of the same situation

Suppose three objects are selected from a container.

- If the order of selection is recorded, an outcome may be a sequence such as (A, C, B) .
- If only the selected objects are recorded, an outcome may be a set such as $\{A, B, C\}$.
- If only object types are recorded, an outcome may be a multiset such as $\{\text{red, red, blue}\}$.

These are different mathematical models and they usually have different numbers of outcomes.

2.3 Why counting models matter

When students make mistakes in combinatorics, the error is often not arithmetic. The error is usually conceptual: they count the wrong type of object. Therefore, before using a formula, one should ask:

1. Are we arranging all objects or selecting only some of them?
2. Does order matter?
3. Are repetitions allowed?
4. Are objects distinguishable?
5. Are rotations or symmetries considered the same?
6. Is the answer obtained directly, or is it easier to count the complement?

3 The Product Rule

3.1 Statement of the rule

Product rule

Suppose a process consists of several consecutive stages. If the first stage can be done in a_1 ways, the second in a_2 ways, and so on, and if the number of choices at each stage is known, then the total number of possible complete outcomes is

$$a_1 a_2 \cdots a_m.$$

The product rule is the foundation of most counting formulas. Factorials, permutations, ordered selections, and sequences with repetition can all be derived from it.

3.2 Sequential construction

The product rule applies when an outcome can be constructed step by step. Each complete outcome corresponds to one path through the stages.

Example: meal choices

A cafeteria offers 3 soups, 5 main dishes, and 4 desserts. A full meal consists of one soup, one main dish, and one dessert. The number of possible meals is

$$3 \cdot 5 \cdot 4 = 60.$$

The outcome is an ordered triple

(soup, main dish, dessert).

3.3 Changing numbers of choices

The number of available choices may change from stage to stage. This is especially important when repetitions are not allowed.

Example: unique access badge

A badge code consists of three different letters chosen from 8 available letters. Since no letter may repeat, the number of possibilities is

$$8 \cdot 7 \cdot 6 = 336.$$

After the first letter is chosen, only 7 remain. After the second, only 6 remain.

4 The Sum Rule

4.1 Statement of the rule

Sum rule

If a task can be completed in one of several disjoint cases, and the cases contain $b_1, b_2, \dots, b_m$ possibilities, then the total number of possibilities is

$$b_1 + b_2 + \dots + b_m.$$

The cases must be disjoint. This means that no outcome may be counted in more than one case.

Double counting

If cases overlap, simple addition counts some outcomes more than once. In that situation, the cases must be refined into disjoint cases or corrected using inclusion-exclusion.

4.2 Splitting into cases

Example: codes ending with a special symbol

A code consists of two letters followed by one digit. Letters are chosen from 6 letters, digits from 10 digits. Suppose the code is considered special if it ends in 0 or 5.

There are two disjoint cases:

last digit is 0 or last digit is 5.

Each case gives $6 \cdot 6 = 36$ possibilities, so the total is

$$36 + 36 = 72.$$

Equivalently, one may write $6 \cdot 6 \cdot 2 = 72$.

5 Factorials

5.1 Definition

Factorial

For a positive integer n , the factorial $n!$ is defined by

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1.$$

By convention,

$$0! = 1.$$

The convention $0! = 1$ is not arbitrary. It makes formulas work correctly when no choices remain. There is exactly one way to arrange nothing: do nothing.

5.2 Factorials as products of decreasing choices

Factorials naturally appear when arranging distinct objects without repetition.

Example: arranging paintings

Five different paintings are to be placed in a row. There are 5 choices for the first position, 4 for the second, and so on. Thus the number of arrangements is

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120.$$

6 Permutations

6.1 Definition and formula

Permutation

A permutation of n distinct objects is an ordering of all n objects.
The number of permutations of n distinct objects is

$$n!.$$

Permutations are used when:

- all objects are used,
- order matters,
- objects are distinct,
- repetitions are impossible because every object is used once.

6.2 Structural interpretation

A permutation can be viewed as a bijection between positions and objects. If there are n positions and n objects, then a permutation assigns exactly one object to each position.

position	1	2	$\cdots$	n
object	$a_{\sigma(1)}$	$a_{\sigma(2)}$	$\cdots$	$a_{\sigma(n)}$

Here σ describes the ordering.

6.3 Blocks in permutation problems

Sometimes a condition forces several objects to remain together. Then they can be temporarily treated as a single block.

Example: two objects kept together

Six distinct files are arranged in a folder sequence. Suppose two particular files must remain next to each other. Treat the two files as one block. Then there are 5 units to arrange: the block plus the other 4 files. These can be arranged in $5!$ ways. Inside the block, the two files can be ordered in $2!$ ways. Hence the total number is

$$5! \cdot 2!.$$

6.4 Counting by complement

If a condition says that something must not happen, it is often easier to count all arrangements and subtract the forbidden ones.

Complement method

If A is the set of acceptable outcomes and B is the set of forbidden outcomes inside a finite universe Ω , then

$$|A| = |\Omega| - |B|.$$

This is useful when the forbidden condition is easier to count than the allowed condition.

7 k-Permutations: Ordered Selections Without Repetition

7.1 Definition

k-permutation

A k -permutation of n distinct objects is an ordered selection of k objects from n , without repetition.

The number of such selections is

$$P(n, k) = n(n-1)(n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

This model is used when:

- only k objects are chosen from n ,
- order matters,
- the same object cannot be chosen twice.

7.2 Why the formula works

There are k positions to fill. The first position can be filled in n ways, the second in $n-1$ ways, and so on. Therefore,

$$P(n, k) = n(n-1) \cdots (n-k+1).$$

Example: assigning roles

A project group has 9 members. Three different roles must be assigned: coordinator, reviewer, and presenter. One person cannot take two roles. The number of assignments is

$$9 \cdot 8 \cdot 7 = P(9, 3) = \frac{9!}{6!} = 504.$$

Order matters because the roles are different.

7.3 Relation to permutations

A full permutation is a special case of a k -permutation with $k = n$:

$$P(n, n) = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!.$$

8 Sequences with Repetition

8.1 Definition

Sequence with repetition

A sequence of length k over an alphabet of n symbols is an ordered list of k symbols where each position may contain any of the n symbols.

The number of such sequences is

$$n^k.$$

This model is used when:

- order matters,
- repetitions are allowed,
- each position is chosen from the same set of possibilities.

8.2 Independent positions

The formula n^k follows from the product rule:

$$\underbrace{n \cdot n \cdot \dots \cdot n}_{k \text{ factors}} = n^k.$$

Example: sensor labels

A device label consists of 4 symbols, each chosen from the set $\{A, B, C, D, E, F\}$. Symbols may repeat. The number of labels is

$$6^4 = 1296.$$

8.3 Sequences versus numbers

A sequence is not always the same as a number. For example, the sequence 0073 is a valid four-symbol code, but it is not a four-digit number in the usual sense because a number cannot begin with leading zeros if it is described as a four-digit number.

Codes and numbers

A code may begin with zero; a usual multi-digit number may not. Therefore, “four-digit code” and “four-digit number” are usually different counting problems.

9 Combinations

9.1 Definition

Combination

A combination is an unordered selection of k objects from n distinct objects, without repetition.

The number of combinations is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

This model is used when:

- only k objects are chosen from n ,
- order does not matter,
- objects are distinct,
- each object can be chosen at most once.

9.2 Deriving the formula

First count ordered selections without repetition:

$$P(n, k) = \frac{n!}{(n - k)!}.$$

Each unordered set of k selected objects can be ordered in $k!$ ways. Therefore, the number of unordered selections is

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k!(n - k)!}.$$

9.3 Symmetry of binomial coefficients

Symmetry

For $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n}{n - k}.$$

Choosing k objects to include is equivalent to choosing $n - k$ objects to exclude.

Example: selecting books

A library display contains 12 different books. We want to select 5 of them for a shelf. Since the order on the shelf is not being recorded, the number of possible selections is

$$\binom{12}{5} = 792.$$

If the order on the shelf were important, this would become a k -permutation problem instead.

10 Permutations with Repeated Elements

10.1 Indistinguishable copies

Sometimes not all objects are individually distinguishable. If a collection contains repeated types, then simply using $n!$ overcounts because exchanging identical objects does not create a new observable arrangement.

Permutation with repeated elements

Suppose n objects consist of r types, with

$$n_1 + n_2 + \cdots + n_r = n.$$

Objects of the same type are indistinguishable. The number of distinct linear arrangements is

$$\frac{n!}{n_1!n_2!\cdots n_r!}.$$

10.2 Why division appears

If all n objects were labeled, there would be $n!$ arrangements. But within each group of identical objects, rearrangements do not change the visible arrangement. Therefore we divide by the factorials of the multiplicities.

Example: colored tiles

A row is made from 8 tiles: 3 red, 3 blue, and 2 yellow. Tiles of the same color are indistinguishable. The number of color patterns is

$$\frac{8!}{3!3!2!} = 560.$$

10.3 Connection with multinomial coefficients

The expression

$$\frac{n!}{n_1!n_2!\cdots n_r!}$$

is called a multinomial coefficient. It counts the number of ways to distribute n positions among r categories of prescribed sizes.

11 Circular Arrangements**11.1 Rotational equivalence**

In a linear arrangement, positions are fixed: first, second, third, and so on. In a circular arrangement, rotations may represent the same arrangement. This changes the count.

Circular permutation

If n distinct objects are arranged around a circle and arrangements that differ only by rotation are considered the same, then the number of circular arrangements is

$$(n-1)!.$$

11.2 Why the formula is $(n - 1)!$

In a linear arrangement there are $n!$ orders. Around a circle, each circular arrangement can be rotated in n ways without changing the relative order. Therefore,

$$\frac{n!}{n} = (n - 1)!.$$

Equivalently, fix one object as a reference point, and arrange the remaining $n - 1$ objects around it.

Example: circular display

Seven different icons are placed around a circular dial. Only relative circular order matters. The number of different arrangements is

$$(7 - 1)! = 6! = 720.$$

11.3 Reflections

Sometimes mirror images are also considered the same, for example in necklaces or bracelets. If rotations and reflections are both identified, the formula may change. For $n \geq 3$ distinct objects, the number of bracelets up to rotation and reflection is

$$\frac{(n - 1)!}{2}.$$

This model is different from ordinary circular seating.

Circle versus bracelet

A circular seating arrangement usually treats clockwise and counterclockwise orders as different, because left and right neighbors are physically different. A bracelet often treats mirror images as the same. Always check which symmetries are being identified.

12 Distinguishable and Indistinguishable Objects

12.1 Distinguishable objects

Objects are distinguishable if they are treated as separate individuals. For example, three devices may be labeled D_1, D_2, D_3 . Even if they are physically similar, the labels make them distinct in the mathematical model.

12.2 Indistinguishable objects

Objects are indistinguishable if only their type is recorded. For example, three identical black tokens may all be represented by B . Exchanging two black tokens does not create a new outcome.

Example: two modeling choices

A bag contains 2 red tokens and 2 green tokens.

If all tokens are labeled, a linear arrangement of all tokens has

$$4! = 24$$

possibilities.

If only colors are recorded, the number of visible color arrangements is

$$\frac{4!}{2!2!} = 6.$$

The physical bag may be the same, but the mathematical model is different.

13 Order: Recorded or Ignored

13.1 Ordered outcomes

When order is recorded, outcomes are sequences. For example,

$$(A, B, C) \neq (C, B, A).$$

This is typical when positions, time order, ranks, roles, or stages are important.

13.2 Unordered outcomes

When order is ignored, outcomes are sets or multisets. For example,

$$\{A, B, C\} = \{C, B, A\}.$$

This is typical when only membership matters.

Diagnostic question

Ask: if I swap the order of two selected objects, do I get a genuinely different outcome?

If yes, order matters. If no, order does not matter.

14 A Decision Map for Counting Models

The following decision map summarizes the main choices without tying them to any particular exercise.

Decision procedure

1. First decide whether all objects are used or only some objects are selected.
2. If all objects are used, decide whether the arrangement is linear, circular, or affected by repeated indistinguishable types.
3. If only some objects are selected, decide whether order matters.
4. If order matters, decide whether repetition is allowed.
5. After the model is chosen, handle restrictions by direct counting, complements, or case splitting.

Question	Answer	Model
Are all objects used and arranged in a line?	Yes; all objects are distinct.	Permutation: $n!$
Are all objects used around a circle?	Yes; rotations are identified.	Circular permutation: $(n - 1)!$
Are all objects used but some are identical?	Yes; only types are visible.	Repeated elements: $\frac{n!}{n_1! \cdots n_r!}$
Are only some objects selected and order is ignored?	Yes; no repetition.	Combination: $\binom{n}{k}$
Are only some objects selected and order matters?	Yes; no repetition.	k -permutation: $\frac{n!}{(n-k)!}$
Are positions filled from the same set and repetition is allowed?	Yes; every position has n choices.	Sequence: n^k

15 Counting with Restrictions

Many problems add restrictions: a position must contain a specific type, two objects must be adjacent, a repeated symbol must appear, or a selected group must contain at least one object from a category.

There is no single formula for restrictions. Instead, one combines the basic counting rules.

15.1 Direct counting

Direct counting means building only the acceptable outcomes.

Example: first symbol fixed

A six-symbol code is built from 10 digits, repetition allowed, but the first symbol must be 7. The first position has 1 choice; each of the remaining five positions has 10 choices. Thus the number of codes is

$$1 \cdot 10^5 = 100000.$$

15.2 Complement counting

Complement counting is useful for phrases such as “at least one”, “not all”, or “contains a repetition”.

Example: at least one special character

A password has 4 characters chosen from 20 symbols. Repetition is allowed. Five symbols are special. How many passwords contain at least one special symbol?

All passwords:

$$20^4.$$

Passwords with no special symbols:

$$15^4.$$

Therefore passwords with at least one special symbol:

$$20^4 - 15^4.$$

15.3 Case splitting

When outcomes naturally fall into disjoint categories, split the problem into cases and use the sum rule.

Example: selecting from categories

A panel of 4 people is chosen from 7 analysts and 5 engineers. How many panels contain exactly 1 engineer or exactly 2 engineers?

These cases are disjoint.

$$\binom{5}{1} \binom{7}{3} + \binom{5}{2} \binom{7}{2}.$$

The first term counts panels with exactly 1 engineer; the second counts panels with exactly 2 engineers.

16 Counting and Event Probabilities

16.1 Counting favorable outcomes

In a uniform finite probability space,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

Counting methods are used twice: first to count the whole sample space, then to count the event.

Example: choosing a subset uniformly

From a group of 10 distinct items, 3 are selected uniformly at random. What is the probability that a fixed item is selected?

The sample space consists of all 3-element subsets:

$$|\Omega| = \binom{10}{3}.$$

Favorable subsets contain the fixed item, so the other 2 selected items must be chosen from the remaining 9:

$$|A| = \binom{9}{2}.$$

Thus

$$\mathbb{P}(A) = \frac{\binom{9}{2}}{\binom{10}{3}} = \frac{3}{10}.$$

16.2 Ordered versus unordered probability spaces

The sample space must match the random mechanism.

Example: ordered draw versus unordered hand

If three cards are drawn one after another and order is recorded, the sample space consists of ordered triples. If three cards are drawn and only the hand is recorded, the sample space consists of 3-element subsets.

Both models can describe related experiments, but they are not the same sample space.

Often, if the probability model is uniform in both descriptions and the event is order-invariant, either model may give the same probability. But this should not be assumed blindly. The model must be chosen consciously.

17 Inclusion-Exclusion

17.1 Two sets

When two counted sets overlap, the union is not counted by simple addition.

Inclusion-exclusion for two sets

For finite sets A and B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The term $|A \cap B|$ is subtracted because it was counted twice in $|A| + |B|$.

Example: language skills

In a group, 18 people know Python, 12 know SQL, and 7 know both. The number of people who know Python or SQL is

$$18 + 12 - 7 = 23.$$

17.2 Three sets

For three finite sets,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

This formula alternately adds and subtracts intersections to correct overcounting.

18 Binomial Coefficients and Algebra

18.1 Pascal identity

Pascal identity

For integers $1 \leq k \leq n-1$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Combinatorial interpretation: to choose k objects from n , fix one distinguished object. Either it is not chosen, giving $\binom{n-1}{k}$, or it is chosen, giving $\binom{n-1}{k-1}$.

18.2 Binomial theorem

Binomial theorem

For every nonnegative integer n ,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

The coefficient $\binom{n}{k}$ appears because, when multiplying $(x+y)(x+y) \cdots (x+y)$, one must choose which k factors contribute y ; the remaining $n-k$ factors contribute x .

19 Common Counting Models: Summary Table

Model	Outcome type	Conditions	Count
Permutation	linear arrangement of all objects	all used, order matters, distinct	$n!$
Circular permutation	circular arrangement	rotations identified	$(n - 1)!$
Permutation with repeated elements	arrangement of a multiset	repeated types indistinguishable	$\frac{n!}{n_1! \cdots n_r!}$
Combination	unordered selection	choose k , no repetition	$\binom{n}{k}$
k -permutation	ordered selection	choose k , no repetition	$\frac{n!}{(n-k)!}$
Sequence with repetition	ordered sequence	length k , repetition allowed	n^k

20 Worked Conceptual Examples

The following examples are not solutions to a task list. They illustrate how one identifies structures.

20.1 Example 1: arranging distinct speakers

Four different speakers give short talks in one session. The order of talks matters and all speakers appear exactly once. This is a permutation problem:

$$4! = 24.$$

20.2 Example 2: choosing speakers without roles

Four speakers are chosen from 11 candidates, but no order or role is assigned. This is a combination problem:

$$\binom{11}{4} = 330.$$

20.3 Example 3: choosing speakers with roles

From 11 candidates, one chair, one discussant, and one presenter are chosen. Roles are different, so order matters in the mathematical sense. This is a k -permutation:

$$P(11, 3) = 11 \cdot 10 \cdot 9 = 990.$$

20.4 Example 4: repeated symbols in a code

A system generates codes of length 6 from the symbols A, B, C, D . Repetition is allowed. This is a sequence with repetition:

$$4^6.$$

20.5 Example 5: color pattern with repeated elements

A pattern has 10 positions: 4 black, 4 white, and 2 gray. Only colors are visible. The number of patterns is

$$\frac{10!}{4!4!2!}.$$

20.6 Example 6: probability from counting

A 5-person team is selected uniformly from 14 people. What is the probability that two fixed people are both selected?

All teams:

$$\binom{14}{5}.$$

Teams containing both fixed people: choose the remaining 3 team members from the other 12 people:

$$\binom{12}{3}.$$

Thus

$$\mathbb{P} = \frac{\binom{12}{3}}{\binom{14}{5}}.$$

21 Frequent Mistakes

Mistake 1: using combinations when order matters

If different positions, ranks, roles, or time order are recorded, then (A, B) and (B, A) may be different. In that case, combinations are not appropriate.

Mistake 2: using permutations when only a subset is selected

A permutation $n!$ arranges all n objects. If only k objects are selected, use a combination or a k -permutation depending on whether order matters.

Mistake 3: ignoring repeated types

If objects of the same type are indistinguishable, using $n!$ overcounts. Divide by the factorials of repeated multiplicities.

Mistake 4: adding overlapping cases

The sum rule only applies directly to disjoint cases. If cases overlap, use disjoint case splitting or inclusion-exclusion.

Mistake 5: confusing physical process with mathematical recording

The same physical draw may be modeled as an ordered sequence, an unordered set, or a multiset of types. The chosen recording rule determines the sample space.

22 Checklist for Solving Counting Problems

Before doing arithmetic, answer the following questions in writing:

1. What is one elementary outcome?
2. Is the outcome a sequence, a set, a multiset, a linear arrangement, or a circular arrangement?
3. Are all objects used or only some?
4. Does order matter?
5. Are repetitions allowed?
6. Are objects distinguishable or indistinguishable?
7. Are there restrictions that require case splitting?
8. Is it easier to count the complement?
9. Are the cases disjoint?
10. If this is a probability problem, are elementary outcomes equally likely?

23 Conceptual Summary

Combinatorics for probability is the art of modeling finite sample spaces correctly. The formulas are short, but the modeling decisions are subtle.

- Use the product rule when outcomes are built in stages.
- Use the sum rule when outcomes are split into disjoint cases.
- Use permutations when all distinct objects are arranged linearly.
- Use circular permutations when only relative circular order matters.
- Use combinations when selecting without order.
- Use k -permutations when selecting with order and without repetition.
- Use sequences with repetition when each position can be filled independently.
- Use permutations with repeated elements when arranging indistinguishable copies of several types.
- Use complements and inclusion-exclusion when restrictions are easier to count indirectly.

The essential habit is to describe the outcome before choosing the formula. Once the outcome type is mathematically clear, the counting method becomes much less mysterious.

A Formula Sheet

Structure	Formula
Product rule	$a_1 a_2 \cdots a_m$
Sum rule for disjoint cases	$b_1 + b_2 + \cdots + b_m$
Permutation	$n!$
Circular permutation	$(n - 1)!$
k -permutation	$P(n, k) = \frac{n!}{(n - k)!}$
Sequence with repetition	n^k
Combination	$\binom{n}{k} = \frac{n!}{k!(n - k)!}$
Permutation with repeated elements	$\frac{n!}{n_1! n_2! \cdots n_r!}$
Two-set inclusion-exclusion	$ A \cup B = A + B - A \cap B $
Classical probability	$\mathbb{P}(A) = \frac{ A }{ \Omega }$

B Recommended Learning Path

A good way to study this topic is not to memorize formulas first, but to practice classifying outcome types.

1. Start from product and sum rules.
2. Learn to recognize whether order matters.
3. Distinguish selection from arrangement.
4. Study repetition and distinguishability.
5. Only then memorize the standard formulas.
6. Finally, apply counting to probability by identifying Ω and the event A .